

Sajed Zarinpour Nashroudkoli

Cover Letter

Dear Dr. George D. Cresswell,

I am writing to express my strong interest in the PhD position in 'Computational Cancer Genomics/Bioinformatics' in your group at the St. Anna Children's Cancer Research Institute. With a master's degree in Numerical Analysis and a solid background in applied mathematics, data analysis, and machine learning, I am eager to contribute to your research on the evolution and mutational processes of pediatric cancers.

I completed my master's degree in Numerical Analysis at the Institute for Advanced Studies in Basic Sciences, where I developed a strong foundation in biophysical and mathematical modeling. Through the joint winter school with Graz University (as mentioned in my CV) and my own research on integrating imaging techniques to construct 3D breast cancer models, I became deeply interested in applying theoretical models to clinical problems. My master's thesis focused on employing neural networks to solve partial differential equations, highlighting my interest in bridging applied mathematics with practical medical applications. Later, I directed my attention toward cancer modeling, studying works such as "Spatio-genetic and phenotypic modelling elucidates resistance and re-sensitisation to treatment in heterogeneous melanoma", which presents a cutting-edge mathematical framework for investigating drug resistance. Its focus on mutational processes and tumor evolution strongly resonated with the themes emphasized in your group's research, particularly in understanding which children are most at risk of poor treatment response.

As I explored your research, I was particularly drawn to your papers "Model-based tumor subclonal reconstruction" and "Exploiting evolutionary steering to induce collateral drug sensitivity in cancer". I find your application of Physics-Informed Neural Networks (PINNs) especially compelling, as they achieve more reliable and interpretable results compared to purely data-driven approaches. I am also deeply impressed by how you integrate tumor evolution into your models and rigorously quantify data uncertainty. Moreover, I greatly admire the experimental design in your studies on drug resistance, especially your consideration of both surviving and deceased cell populations and your hypothesis connecting these dynamics to the evolutionary history of the tumor. This comprehensive and mechanistic approach aligns closely with my own research philosophy.

I believe my background in applied mathematics, data analysis, and machine learning (particularly in modeling cancer biology) makes me a strong candidate for this position. My professional experience as a software developer has strengthened my programming, teamwork, and problem-solving skills, and taught me to perform effectively under pressure and tight deadlines. Beyond the technical aspects, my motivation to study cancer biology is deeply personal, having lost my aunt to breast cancer. I am eager to translate my skills and passion into developing computational methods that predict treatment response and improve patient outcomes. My research interests include oncology, biophysical modeling, physics-informed machine learning, uncertainty quantification, and time-series data analysis. Ultimately, I aspire to pursue an academic career where I can combine computational modeling with biological insight to advance cancer research.

Thank you very much for considering my application. I would be honored to contribute to your group and further discuss how my background and interests align with your research goals.

Sincerely,

Sajed Zarinpour Nashroudkoli